A Time of Their Own: Agent-Held Temporal Evidence for Persistent AI Agents

A Perspective

Ioan Daniel Ionici

Independent Researcher, Oradea, Romania

ORCID: https://orcid.org/0009-0009-8183-3656

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Status	Draft of version 1.4 for review, 9 November 2026 (not for citation)
Changes	Version 1.4 reports the outcome of a second study, which tested Level 2 for its cheapest candidates and found them reconstructed from free optimizer telemetry [24]; it updates the description of that level (§4, Table 2), adds to the closure rule of §5 an exit for candidates that are not identifiable, and adds two cautions to §6. Version 1.3 makes the closure rule of §5 operational, states that the accompanying study does not test the programme and what carries over from it, and adds a statement of priority against rollback protection, machine unlearning, and training-data attribution [8, 28, 46]. Version 1.2 cites prior work that used the cumulative norm of parameter updates as a model clock [18], and prior work on safe interruptibility, corrigibility, state continuity, and learning progress [16, 17, 27, 42, 44, 64], and states the storage cost of the experiential index more precisely. Version 1.1 cited prior and related work omitted from version 1.0 [33, 53, 54, 62]. The hypothesis is unchanged in every version; version 1.3 narrows the claims about the accompanying study and about priority, and makes the closure rule stricter; version 1.4 records that the cheapest candidates of the experiential index are not identifiable in the regimes tested
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Relation	Accompanies two closed studies, whose empirical claims it does not extend: an exploratory study [25], whose design principles for evaluation it uses, and a study of Level 2 [24], whose negative result for the candidates tested it reports (§4, §5)
Scope	Functional properties of persistent artificial agents. No claim is made about consciousness, moral status, or personal identity
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Abstract

Persistent artificial agents are ordinarily anchored in time from outside, and whatever timing they learn internally remains part of a state that can be saved and restored. When that state is restored, every internal clock and record is restored with it: from its restored state alone, an agent cannot tell whether it has continued without interruption or been rolled back. Persistence, continuity, and identity of such agents have recently been recognised as open problems, and mechanisms that supply evidence of another kind exist elsewhere — in hardware security, cryptography, and the neuroscience of timing — but they serve the platform, or the guard checks of protected modules, not an agent's predictive model of itself. We classify the temporal evidence available to an agent — of order, duration, sequential computation, continuity, and its own change — by who can write, reset, forge, or copy it and by whether its self-model can read it, and we distinguish continuity claims adjudicated by the platform from raw evidence interpreted by the agent, treating their equivalence as a legitimate negative result. Our hypothesis is conditional: knowing

how old one's knowledge is, detecting suspension, restoration, or copying, and ordering, revoking, and anticipating one's own changes benefit from authenticated evidence that is not restored with the agent's state, chiefly when external records are absent, delayed, or untrusted; with complete, trusted logs, replay reconstructs any deterministic agent, and an internal alternative can only be cheaper. Physical irreversibility is one route to such evidence, not the argument for it. We separate two roles: a non-restorable time reference with a state-bound operation counter detects and distinguishes suspension and restoration, whereas an index of accumulated experience-driven change, read against that reference, serves the agent's model of its own adaptation. A completed exploratory study supplies design principles for this evaluation — information-matched controls, competence audits, and equivalence margins — and shows why local update statistics must be excluded before traces are read as accumulated history; it does not test the programme. A second study tested the cheapest candidates for the index of accumulated change, including the update-norm clock used to schedule replay in continual learning of language models: in small models trained with Adam, each was reconstructed from free optimizer telemetry, and as a replay clock the update-norm index did no better than a free one. The index remains a definition; its cheap candidates are not identifiable in the regimes tested, and the continuity anchor is untested. Each proposed test is paired with a conventional competitor that can close it, and access to evidence of restoration confers no control over it. We make no claim about consciousness or moral status.

Keywords: agent-held temporal evidence; persistent agents; self-models; state continuity; rollback; irreversibility; time perception; provenance; hardware roots of trust.

What this perspective offers

- A classification of the temporal evidence available to an agent — of order, duration, sequential computation, continuity, and its own change — by who can write, reset, forge, or copy it, and by whether the agent's self-model can read it, under explicit threat models (Table 1).
- An elementary result, given a machine-checked form in earlier work [54], that locates the problem: an agent whose complete restorable state is restored cannot detect the restoration from that state alone; detection requires authenticated evidence that was not restored with it and to which the restored state is bound (§3).
- A separation of two roles: a continuity anchor — a non-restorable time reference and a state-bound operation counter — detects and distinguishes suspension and restoration, whereas a dual clock relating accumulated experience-driven change to that reference serves the agent's model of its own adaptation (§4).
- A sequence of tests, each with the conventional mechanism that could make it unnecessary (Table 2), self-contained definitions of its methodological gates, and a design principle that separates access to evidence of restoration from control over it.

1. Introduction

Persistent artificial agents are ordinarily anchored in time from outside. Their timestamps, logs, checkpoints, and version identifiers record when things happened, and a language model knows the current date only if the date is written into its input; even the age of its own knowledge is uncertain, because the effective cutoff of what it learned can differ from the reported one [12]. Agents can also learn timing: reinforcement-learning agents develop internal time-keeping when timing matters for the task [48, 49]. But whatever timing an agent learns is part of its

computational state, and that state can be saved, copied, and restored. This becomes a limitation once an agent persists, changes, and must reason about its own past: how long ago it learned something, whether it has been paused, restored, or copied, in what order its own changes occurred, and how an experience will change it.

The limitation is structural. In a system whose whole state can be saved and restored, every internal clock and every internal record is saved and restored with it. Such an agent holds no evidence about elapsed time or about its own change that it could not have been given, reset, or forged by whoever controls its state and inputs. Any count it keeps, any memory it consolidates, and any timestamp it reads can be rolled back together, and after a restoration nothing in the agent's state shows that anything happened.

The concern is not new. Language-model agents have been described as "ontologically stateless", with identifiability, continuity, and persistence undermined despite external memory and tools [50]; it has been argued that such an agent can say the right things about itself without the conditions that would make those statements meaningful co-occurring at the moment of decision, and persistence scores computable from scaffold traces have been proposed [51]; reverting a persistent agent's self-description after its memory has grown does not restore its earlier behaviour [63]; identity for AI agents has been defined as a continuous relationship between what an agent is declared to be and what it is observed to do [43]; a security study of deployed agent systems shows that a faithfully restored checkpoint can still resume an invalid execution, because the state a checkpoint restores differs from the dependencies that secure continuation requires [70]; a survey of always-on agents finds the literature more concerned with accumulating and retrieving state than with governing, recovering, or relinquishing it [15]; and architectures have been proposed that preserve an agent's identity, memory, and code lineage across models, harnesses, and hosts under explicit continuity invariants [73], or that verify governed state transitions with a three-way verdict [21]. Temporal cognition has long been argued to be a key ingredient of intelligent systems [34]; robotic models of episodic and autobiographical memory have been proposed as a synthetic temporal self [52]; and duration can be estimated from accumulated change in a network's activity [57]. Similar intuitions — that AI systems lack a clock or a time of their own — have also been voiced informally; we do not rely on them as evidence.

What this perspective adds is a question about evidence rather than about identity: what can an agent know about elapsed time and about its own change, from which sources, and which of those sources could it not have been given, reset, or forged? The mechanisms that answer this question already exist, in hardware security, in cryptography, and in the neuroscience of timing. Three arrangements must be kept apart: (*a*) continuity evidence held by the platform and used by operators, or by a protected module that checks the freshness of its state and halts on a mismatch [40, 47, 61] — the status quo; (*b*) continuity claims adjudicated by a trusted monitor and passed to the agent as an authenticated result; and (*c*) raw evidence read by the agent's self-model, which infers continuity, duration, and its own change itself. The proposal concerns (*b*) and (*c*), and (*b*) is also the strongest competitor of (*c*). The individual mechanisms are established; the proposed contribution lies in their integration as authenticated temporal evidence available to an agent's self-model, and in the falsifiable programme by which that integration can be evaluated.

Priority should be read narrowly. The mechanisms on which the proposal draws are established or adjacent: trusted counters, freshness checks, and rollback-resistant storage [36, 40, 47, 61];

provenance-guided selective replay [71]; machine unlearning [8]; training-data attribution, which asks which example changed a model offline and from outside it [28, 46]; and the cumulative norm of parameter updates as a model clock [18]. The proposal is not a new clock, cryptographic primitive, unlearning method, or attribution algorithm. Measured against this work, the continuity anchor overlaps substantially with rollback protection, revocation and replay overlap substantially with unlearning and provenance-guided recovery, and source attribution by the agent overlaps partially with training-data attribution. What is plausibly new is the separation of arrangements (*a*), (*b*), and (*c*), the classification of the evidence in Table 1, and a programme in which each test is set against a conventional competitor that can close it.

The motivating intuition is simple to state: an artificial agent needs a connection to a time of its own, and that time must be irreversible. "A time of their own" is used here as shorthand for agent-accessible evidence of elapsed time and continuity, not for subjective temporal experience or for proper time in the sense of relativity; the hypothesis of §2 is the testable form of the intuition. The fields cited throughout supply candidate mechanisms and adjacent formulations of the problem, not convergent evidence for the hypothesis.

This perspective accompanies a closed exploratory study of episode–trace binding [25]. It introduces no data and does not extend that study's claims. What it takes from the study are the design principles of its evaluation, defined in §6 so that they can be used without the study, and two cautions that the study made concrete; the study itself tests none of the levels of §5. A second study, undertaken after version 1.2 and closed before this version, tested Level 2 for its cheapest candidates [24]. Its outcome is reported where that level is described (§4, Table 2), and it adds two cautions to §6.

2. Agent-Held Temporal Evidence

By *temporal evidence* we mean information that bears on five distinct quantities: the *order* of events; the *duration* that has elapsed in physical time; the minimum *sequential computation* that has taken place; the *continuity* of the agent's trajectory — whether it has been interrupted, restored, or forked; and the agent's *trajectory* of change — how experience has altered it. Different mechanisms supply different quantities, and none supplies all of them.

A *self-model* is understood operationally: a component whose outputs are predictions about the agent itself — its future state, the change in its competences, the age of its information, the probability that it has been compromised, or the effect of an update or a rollback. Without such a measurable output, no component counts as a self-model here. Evidence is *agent-held* if it is available to the self-model at decision time, and authenticated where it originates outside the agent's state. It comes in two forms, corresponding to arrangements (*b*) and (*c*) of §1: *adjudicated claims*, in which a trusted monitor evaluates continuity and passes an authenticated result, and *raw agent-bound evidence*, which the self-model reads and interprets itself. Agent-held does not mean stored inside the checkpointable state; the point of the proposal is that some of it must not be.

Five architectural parts are therefore distinguished. The agent's *core state* can be checkpointed, restored, and copied. *Agent-bound evidence sources* lie outside the checkpoint; the agent may read them but cannot write, reset, or decrement them, and the increments and state commitments they

require are executed by a trusted monitor as part of each committed step. The *host orchestrator* controls storage, checkpoints, and restoration; the authority to restore legitimately belongs to the operator, exercised through policies that the trusted monitor enforces and records [9], so that under threat model B the host can execute a restoration but cannot make it authorised. The *trusted reference monitor* lies inside the trusted computing base, mediates the evidence sources, holds the keys with which their readings are authenticated, and executes the protocol-enforced increments. *External channels* carry messages that the agent must validate.

Claims about evidence depend on the threat model. We distinguish three: (*A*) a benign operator and accidental faults — a rollback by mistake, a wrong timestamp, a stale checkpoint, an unrecorded suspension; (*B*) a privileged software adversary — in practice, a compromised host orchestrator — who controls storage, inputs, and checkpoints but cannot alter trusted hardware or forge its attestation; and (*C*) a physical adversary who controls the device itself — its temperature and voltage, its replacement, and the channel through which a physical source is read. Under B, the agent's path to the evidence must itself lie within the trusted computing base — the self-model executes in a trusted execution environment or reads through the monitor — because a host adversary could otherwise replay old readings to a restored agent or alter verification keys stored in the restorable state; freshness rests on that path and on the monitor's own counter, not on nonces stored in the state, which roll back with it. The proposal is plausible under A and B. Under C, no local component guarantees anything by itself. Table 1 classifies the sources of temporal evidence under threat model B.

Table 1. Sources of temporal evidence and their properties (threat model B)

Source	What it measures	Examples	Written by	Reset or restored	Forgeable	Copied with the state	Read by the self-model today
External records	Order; duration, as reported	Timestamps, logs, dates supplied in the input, version identifiers	Platform, operator	Yes	Yes, by whoever controls the channel	Yes	Yes — usually the only source
Internal informational state	Trajectory; order and duration, if learned	Weights, memories, counters, multi-timescale plastic variables [2]	The agent itself	Yes, with any checkpoint	Yes, by whoever can write the state	Yes	Implicitly, as substrate rather than as evidence
Sequential-computation proofs	Minimum sequential computation since a challenge was fixed, not elapsed physical time	Time-lock puzzles, verifiable delay functions [7, 56]	Whoever computes	A proof can be replayed unless its challenge is fresh and fixed outside the state	Lower bound: no, given a fresh challenge; upper bound: yes	Yes	No
Platform roots of trust	Order and continuity	Monotonic counters, state-continuity protocols [36, 40, 47, 61], attestation	Trusted hardware, on increments issued by a trusted monitor	Not by software; by policy where rollback is permitted [9]	Only by breaking the hardware or through a compromised manufacturer	No	No — used by the platform
Non-restorable physical processes	Duration of unpowered intervals (decay clocks); continuity across devices (unclonable functions)	Remanence-decay clocks [55]; device-unique responses [45]	Physics	No, if decay cells are re-armed only by trusted hardware	Not under B; under C, by controlling the environment or the readout	No	No

Under threat model B the adversary controls the platform's storage, inputs, and checkpoints but neither the physics of the device nor the trusted hardware, which moves the trust to its manufacturer. Under threat model C every row may be forged or replayed by whoever controls the device, except that a sequential-computation proof still bounds the work done since a fresh challenge.

Today a self-model reads the first row and, implicitly, the second. The last three rows hold evidence that an adversary of type B cannot have reset — provided that the challenges of sequential proofs are fresh and that decay cells are re-armed only by trusted hardware. Protected modules already read trusted counters at execution time, but for deterministic guard checks that halt on a mismatch [47, 61]; the shift proposed here is to route such evidence, adjudicated or raw, into a learning agent's predictive self-model.

Hypothesis. Some functions of a persistent agent benefit when evidence from the last three rows of Table 1 is agent-held, chiefly when external records are absent, incomplete, delayed, or untrusted. Knowledge age and discontinuity detection rest on those rows directly; ordering and revocation, and anticipation, also require the agent's records of its own trajectory and of the provenance of its changes, anchored in order and time by those rows:

- *Knowledge age*: estimating how much time has passed since something was learned, and hence how likely it is to be stale.
- *Discontinuity detection*: recognising that one's trajectory has been interrupted — by suspension, by restoration to an earlier state, or by forking into copies — or that the timestamps one receives have been altered.
- *Ordering and revocation*: ordering one's own changes without trusted timestamps, and locating and undoing the effect of an experience later found to be compromised [71]; an authorised revocation must be recorded as such, so that it is not mistaken for an adversarial rollback [9].
- *Anticipation*: predicting how, and how durably, an experience will change one's own competences.

What the hypothesis does not say. It does not say that external records are informationally insufficient. With complete and trusted logs, replay reconstructs the state of any deterministic agent; an internal state is then a cache of the log, and its advantage, if any, is computational. The hypothesis concerns regimes — absent, incomplete, delayed, or untrusted records, and limits on compute and latency — not a capacity that logs could never supply. One possible outcome of the programme is therefore *functional separation*: continuity may be best guaranteed by the platform, with the agent modelling only adjudicated claims supplied to it. Raw agent-bound evidence would then be unnecessary, even if temporal self-modelling remained useful.

3. Where Physics Enters

The physical premise behind the motivating intuition is true but proves too much. Every implemented system follows a physical trajectory, and every record is a physical change: erasing information has an irreducible thermodynamic cost [31], and memory records in the direction in which entropy increases [38]. A log entry and a decaying capacitor are equally physical and equally aligned with the arrow of time. That a physical process exists therefore does not distinguish the proposal from the status quo.

What distinguishes it is restorability. Digital computation is engineered to be independent of its physical microstate: memory is refreshed against decay, errors are corrected, and runs can be reproduced bit for bit — as all one hundred training runs of the final iteration of the accompanying study were. The consequence can be stated as an elementary result.

Proposition (rollback indistinguishability). Let an agent's complete restorable state — everything a checkpoint saves — be restored exactly, at time t_1, to its value $S(t_0)$ at an earlier time t_0. Any deterministic function of that state alone then takes the same value as it did at t_0, and for an agent that draws fresh randomness from a memoryless source, the distribution of its outputs is the same as at t_0; no computation over the restored state can therefore distinguish uninterrupted continuation from restoration. Restoration can be detected only by comparing the

restored state with evidence that was not restored with it — a source outside the checkpoint or an authenticated message from outside — and only if the restored state carries an authenticated commitment to an earlier reading of that evidence, or the source holds one to the latest state, and the source and the channel through which it is read can be trusted.

The proposition is elementary, and it is not new. It underlies the state-continuity protocols of protected modules [47, 61]; a machine-checked statement of it for tamper-evident agent memory, under the same name, was published shortly before this article [54], together with a generalisation — no mechanism can decide a property whose deciding fact is absent from its input [53]; and an analogous separation has been proved for the authority of an agent's memory content, which defences based on content or derivation history cannot establish because both can be laundered, and which requires binding to the origin at write time [33]. What the proposition does here is locate the problem. The question is not whether an agent can represent time, which agents can learn to do, but where the evidence it relies on resides. We call a record *irreversible for the agent* if, under the declared checkpoint scope and threat model, neither the agent's own operations nor a restoration of its state can undo it; the term denotes non-restorability relative to those declarations, not irreversibility in an absolute sense. Only authenticated evidence outside the restorable state — held by trusted hardware or produced by a physical process — can be irreversible in this sense, and only under a threat model in which that source and its readout remain trustworthy.

Four properties matter, and physics is one route to them, not the argument for them: *autonomy* — the source evolves without being written by the program; *non-restorability* — no checkpoint restores it; *coupling* — it is read by the computation, ideally by the self-model; and *uniqueness* — it is not copied with the agent. Proposals for mortal computation give up hardware independence and tie what is learned to a particular physical substrate [22, 41]. The motive here is evidence rather than efficiency, but the cost is shared: part of the agent can no longer be copied or replayed.

Biology offers no restorable state, and its timing mechanisms are instructive as analogies and candidate mechanisms, not as evidence that artificial agents need them. The brain has no single central clock: time can be encoded in the evolving state of neural networks [26]; a slowly drifting temporal context supports memory for when things happened [23]; population activity in the lateral entorhinal cortex carries time derived from the flow of experience [65]; and the drift of hippocampal ensembles over days is sufficient to timestamp events [58]. Drift with similar geometry appears in artificial networks trained continually with dropout [1]. The subjective sense of duration has been proposed to draw on the accumulation of bodily, interoceptive states [14, 69] — a debated hypothesis, but the closest biological analogue of a time generated by a system's own physical processes rather than read from an external clock.

4. A Continuity Anchor and a Dual Clock

Two roles must be kept apart: detecting discontinuities of the agent's trajectory, and modelling how the agent changes along it. They need different quantities.

The continuity anchor. Detection requires a non-restorable time reference c_a and a non-restorable count N of the agent's own committed steps, both read through the trusted path and bound to the state: the restored state carries sealed copies of the values at which it was saved,

$c_a(t_0)$ and $N(t_0)$. If N exceeds $N(t_0)$, the agent has operated since that state was saved and has been restored; if N equals $N(t_0)$ while c_a has advanced beyond its calibration error, it has been suspended; if both have advanced together, it has operated without interruption. The pair (c_a, N) is necessary and sufficient for this distinction, and it is what state-continuity protocols already maintain for data [36, 40, 47, 61]. Such protocols remain necessary because, once power is off, the non-volatile state of a trusted execution environment can be reverted undetectably to an older version [27]; the TPM 2.0 clock already separates time references by the events that reset them, with distinct counts of resets and restarts [64]. The sealed reference values must be protected by a key held in trusted hardware, or their digest kept by the monitor, so that a host adversary cannot rewrite them before restoring the state, and commits must be crash-consistent, so that a crash between incrementing the counter and saving the state is not mistaken for a restoration [61].

Protocol sketch. The sufficiency of the pair rests on an ordering that state-continuity protocols already provide; we sketch it only to make the assumptions explicit, not as a new protocol. (1) At each commit the agent proposes a state S_{k+1}; the monitor increments N to N_{k+1}, reads c_a, and seals the tuple of the state digest $H(S_{k+1})$, N_{k+1}, c_a, and a lineage identifier with a key that never leaves trusted hardware. (2) The state and its seal are committed together and crash-consistently, so that a crash leaves either the old pair or the new one. (3) When a state is loaded, the monitor verifies the seal, reads the live values of N and c_a through the trusted path, and classifies the event: an equal counter with the time reference advanced beyond calibration error is a suspension, a larger counter is a restoration, and anything inconsistent with the seal is a forgery. (4) The classification is passed on as an adjudicated claim, or the raw readings as agent-bound evidence. Under threat model B the sketch is sufficient if the host cannot read the sealing key, cannot reorder steps (1) and (2), and cannot interpose between the monitor and the self-model; races between checking and using a reading, and threat model C, lie outside it.

Copying differs from restoration. A fork of the same checkpoint on the same machine shares the time reference and the device-unique responses, and is revealed only when the branches contend for the same counter: once one branch commits, the other finds N ahead of its own record. A clone moved to another device can detect the change of substrate if its state carries a commitment to the original device's unique responses [45], but the instance continuing on the original device sees no change. Neither case reveals which branch is the original or the legitimate successor; only a continuity policy outside the agent can decide that. Device-unique responses are themselves exposed to machine-learning modelling attacks [59] and to hardware Trojans that remain dormant through validation [35], so monotonic counters and attestation in trusted execution environments are the more conservative choice in practice, and unclonable functions a candidate to be tested. Legitimate migration changes the substrate on purpose. It is best represented as an explicit re-anchoring event, authorised by the operator and sealed by the monitor, that binds the old anchor, the new anchor, the state digest, the counter, and the authority under which the move was made; the self-model can then treat the change as an authorised discontinuity with documented continuity rather than as a clandestine copy — the role that fenced migration plays in architectures for runtime-independent agents [73].

The dual clock. The second role — modelling the agent's own adaptation — relates the time reference to an experiential trajectory index c_e, defined as accumulated change along the trajectory, $c_e(t) = \Sigma_{\tau<t}\, D(S(\tau), S(\tau+1))$ over committed steps. The index is non-decreasing and is maintained as one running number; a query about an interval needs only the value stored at its

start. It is called a clock only by analogy: the persistent state itself drifts non-monotonically and can return near earlier regions, which is why the index accumulates change rather than measuring distance from a reference state. The net displacement $D(S(t_0), S(t_1))$ is a separate quantity, which can be small even when much has changed — when an update is learned and later revoked, or when a state cycles. The choice of D is itself an open problem: a distance between weights is not a difference in function. Candidates range from cheap and weak to costly and valid — the norm of parameter change, representational distance on a fixed set of probes, divergence of the action distribution on those probes, change in measured competences, and the estimated counterfactual influence of an experience. The cheapest candidate has already been used: the cumulative norm of parameter updates serves as a model-centric clock for scheduling replay in continual learning of language models [18]. That work compares the clock with step counts only. A second study asked whether such an index carries information beyond logged optimizer statistics [24]. In a synthetic continual-learning world and in a 0.6-billion-parameter language model trained with LoRA and Adam, three parameter-space distances, a functional distance on probes, and the update-norm clock itself were each reconstructed out of sample from cheap telemetry or from competitors of the same cost ($R^2 \geq 0.90$; the exceptions arose only when a schedule family was left out of training). The parameter-space quantities were reconstructed chiefly from the learning rate times the norm of Adam's first moment, which the optimizer already holds, and the functional distance from the change of loss on the same probes. Used as a replay clock, the update-norm index did no better than that free quantity, in a re-implementation and, in aggregate, with the original code. The reason is mechanical: under Adam, the size of a step is set by moments the optimizer keeps. For these candidates, in these regimes, the open question is answered negatively. What remains open are the costly candidates — change in measured competences and counterfactual influence — and larger models, full fine-tuning, and other optimizers. The second study concerns state that changes through gradient updates of weights. Many persistent agents change mainly through writes to a memory outside their weights — records or documents retrieved when needed — or through the state of their tools, where no optimizer keeps a record. For them, an index accumulated over the persistent state, writes and revocations included, is untested; its natural competitor is the count and size of the writes. A candidate is admissible only if it is not reconstructed out of sample from the optimizer's own state — learning rate, moment norms, and schedule — and beats decoders restricted to within-episode update statistics, survives the reset criterion and randomisation of episode order where it is meant to reflect history, predicts a later functional consequence — a change in competence or the effect of an update — better than event counts and logged update summaries, and is used by the agent rather than merely decodable from it.

Read against the time reference, $\rho = \Delta c_e / \Delta c_a$ is the rate at which the agent changes, and the ratio of net displacement to accumulated change measures how directed that change has been. A related internal rate, learning progress — the decrease of prediction error over a window — already drives curiosity-based exploration in developmental robotics [44]; it is measured against the agent's own restorable count of steps, not against a non-restorable reference. Humans judge durations differently in retrospect, from what the interval left in memory, and in prospect, from attention to time [6], and a model that estimates duration from accumulated change in network activity reproduces human biases [57]. These quantities are inputs to anticipation and to the detection of anomalous change, such as an unusually rapid shift after a compromised update. They play no part in detecting suspension or restoration: an agent executing with frozen weights and

no writes to its memory, with $\Delta N > 0$ and $\Delta c_e \approx 0$, is operating, not interrupted, and a restored agent fed synthetic experience before it decides can show normal change while only the anchor reveals the restoration. Knowledge age, finally, is a question for c_a, not c_e: what matters for staleness is how far the world has moved, not how much the agent has experienced.

Engineering limits. Remanence-decay clocks are illustrative candidates for bounded, power-cycle-dependent intervals, not proposed as lifetime clocks. They measure only intervals during which their cells are unpowered, from seconds to hours depending on the hardware, and are sensitive to temperature and to chip-to-chip variation [55]. They do not measure a suspension during which the host stays powered, they saturate long before knowledge age becomes interesting, and re-arming a cell must itself be controlled by trusted hardware, or a host adversary could reset it. A long-horizon reference under threat model B therefore needs a continuously powered timer inside the trusted boundary — and trusted time in such environments is itself exposed to manipulation by the host, which has required dedicated protocols [5] — or an irreversible degradation process, or staggered decay cells re-armed only by trusted hardware. Sequential-computation proofs certify a lower bound on the work done since a fresh challenge, not elapsed time [7]; physical time-lock puzzles, which tie the delay to the evaluation latency of noisy unclonable functions [60], are a further candidate whose physical anchoring brings the risks noted above. And an index read from a learning state inherits every confound of that state, which is where the cautions of §6 apply.

5. Tests and Their Competitors

The hypothesis is testable only against competent conventional alternatives. Table 2 orders the tests from the most basic to the most integrative, credits the mechanisms on which each builds, and names the conventional competitor that could make it unnecessary. The order is not a chain of presuppositions: each level can be closed independently, and if an earlier level is closed, later levels proceed with the winning conventional mechanism substituted for the one that lost. Closing a physical implementation does not close the corresponding functional question. A level is closed — and the corresponding part of the hypothesis abandoned for that regime — if its competitor is non-inferior on the level's primary metric within a margin fixed in advance, the smallest difference that would matter [30], and is no worse on any cost dimension declared in advance: hardware, energy, latency, storage, communication, recomputation, error rates, availability, robustness to attack, and administrative complexity. A result in which the competitor is better on some dimensions and worse on others is reported as unresolved, with the trade-off stated; it neither closes the level nor counts as support for the hypothesis. A test can also end before any comparison on the primary metric. If a candidate is reconstructed out of sample from the competitor's own inputs, above a threshold fixed in advance, it carries no testable information beyond them. The hypothesis is then reported as not identifiable for that candidate and regime, never as equivalence and never as support. This exit was added in version 1.4, after the second study met it [24]. Each level also declares a primary metric: calibration error and energy over the measured interval for Level 1; predictive validity per unit of storage for Level 2; recomputation for Level 3; detection latency and false-alarm rate for Level 4; decision regret for Level 5; and prediction error for Level 6.

These rules become a test only through an evaluation contract for each level, fixed before any data are collected: the benchmark environment, the threat model, the competitor, the primary metric, the declared cost dimensions, the margin, the planned sample size or simulation budget, and the statistical decision procedure. No such contract is provided here. This article proposes falsifiable questions and candidate metrics, not complete protocols for falsifying them.

Table 2. Tests, the work they build on, and the competitors that can close them

Level	Question	Builds on	Competitor that can close it	Gate from the method of §6
1. Time reference	Can a non-restorable source give the agent calibrated evidence of elapsed time?	Remanence-decay clocks [55]; trusted time [5]; sequential-computation proofs [7, 56]; physical time-lock puzzles [60]	A trusted external clock where a trusted channel exists; a verifiable delay function where it does not; a continuously powered trusted timer	Calibration against an independent reference; a matched informational simulation of the same signal
2. Experiential index	Does accumulated change of the persistent state, under a chosen distance, predict the agent's own adaptation?	Time from network change; drift as a timestamp; update magnitude as a replay clock; per-parameter importance accumulated along the trajectory, which approximates a gradient statistic [1, 3, 18, 57, 58, 72]	Event counts; the optimizer's own state (learning rate, moment norms, schedule), available at negligible cost; update norms or policy divergence logged in checkpoint metadata; for state that changes without gradient updates, the count and size of writes	Must not be reconstructed out of sample from the optimizer's state, and must beat decoders restricted to within-episode update statistics by a prespecified margin
3. History	Do attribution and self-prediction require accumulated history?	Provenance and selective replay; episode–trace binding [25, 71]	The log with replay; a platform provenance index; the agent with its state reset, as negative control	Reset criterion: intact history beats reset by a prespecified margin
4. Discontinuity	Can the agent detect suspension, restoration, forking, and altered timestamps, and use graded evidence of them?	State continuity, authorised rollback, checkpoint fencing, branch primitives [9, 36, 40, 47, 61, 67, 74]	A local trusted monitor that computes the continuity predicate and passes an authenticated flag; a verifier of governed transitions [21]	Equivalence with a complete flag as competence audit; comparison with the flag delayed, incomplete, or absent
5. Closed-loop use	Does agent-held evidence improve decisions, such as re-verifying stale knowledge or revoking compromised updates?	Knowledge cutoffs; memory decay computed from timestamps [12, 75]	Timestamps, logs, freshness metadata, and authenticated agent telemetry [20]	Competence audit of the conventional agent
6. Temporal self	Does the agent predict when and how it changed, and how it will change?	Synthetic temporal self; persistence scores [51, 52]	The current state plus a description of the experience	Onset versus skill; equivalence tests

Level 2 has been tested for its cheapest candidates since version 1.2: they are not identifiable in the regimes tested [24] (§4).

Level 4 is where the proposal is most distinctive and most exposed. The strongest competitor is local: a trusted monitor on the same host that evaluates the continuity predicate when a state is loaded and passes an authenticated flag. It shares the root of trust and the readout path of raw

agent-bound evidence and avoids the calibration error of decay clocks, so equivalence, or an advantage for the flag, is the expected result for detection itself. Equivalence with a complete flag therefore serves as the competence audit of the comparison, and the level is closed for detection if the flag is non-inferior to the agent at no greater cost, by the rule above, across the regimes the tests must vary — a flag that is delayed, incomplete, or absent, as on a disconnected device or under an orchestrator that withholds it. What would then remain is whether graded, raw evidence improves the agent's downstream self-predictions beyond a binary flag, which Levels 5 and 6 test. Recent systems resolve the side-effects of rollback and the reconciliation of branches entirely within the platform — by fencing irreversible tool effects across checkpoint restoration [74] and by operating-system primitives for forking and committing exploratory branches [67] — and they belong to the competitor.

Example tasks for Levels 4–6. Four tasks make the later levels concrete; each compares raw agent-bound evidence with an adjudicated flag, with timestamps and logs, and with no evidence at all. (*i*) *Re-verification of stale knowledge*: the agent holds facts that change in the world at different rates and, after an interval, must decide which to use and which to re-verify; the metrics are decision regret, the rates of needless re-verification and of stale use, the cost of verification, and calibration. (*ii*) *Revocation of a compromised experience*: an episode is declared compromised after it has changed the persistent state, and the agent must decide what to restore, replay, and re-evaluate; the metrics are residual influence, distance to an agent that never saw the episode, collateral loss, recomputation, and false revocations, against full reset and provenance-guided selective replay [71]. (*iii*) *Prospective update*: before accepting an update, the agent predicts which competences it will change, how strongly, and for how long; the metrics are prediction error, Brier scores for the risks predicted, and damage to protected competences. (*iv*) *Continuity under a delayed flag*: the agent operates after a restoration before an external verdict arrives; the metrics are detection latency, wrong decisions before detection, cumulative regret, and false alarms. Evaluation contracts that score state mutation and recovery obligations rather than answers [15], and the failure modes of checkpoint restoration documented in deployed agent systems [70], supply adversarial scenarios for all four.

6. Method and Cautions from Two Completed Studies

The accompanying study asked a narrow question — whether binding an episode record to a summary of the internal change the episode produced helps a model attribute the episode's source — in an offline, synthetic setting whose predictor never drives behaviour [25]. It does not test the programme: physical timing, continuity detection, delayed flags, closed-loop decisions, and self-prediction are absent from its data. What carries over are its gates, defined here so that they can be used without that study; each level of Table 2 still needs its own evaluation contract (§5), with controls specific to its threat model. An *information-matched control* receives the same evidence in another form, so that a comparison tests how the evidence is represented, not how much of it is available. The *decoding ceiling* of a level is the performance of an independent decoder, trained on substantially more data than the agents, in predicting the level's target from the evidence available; it estimates what the evidence supports. A control passes the *competence audit* if it reaches a prespecified fraction of that ceiling — one half in the accompanying study — and exceeds a version of itself deprived of the evidence by a prespecified margin. *Equivalence* is declared when the confidence interval of a difference lies within a prespecified margin, by two

one-sided tests [30]. *Onset versus skill* reports separately whether, and when, each model first exceeds a prespecified minimum skill, and the skill both reach once both have exceeded it, so that a difference in whether learning begins is not mistaken for a difference in what is learned. *Construct-validity diagnostics* retrain the decoder on reduced inputs to check what actually carries the signal. Two results of the study are cautions for the present proposal.

- **An apparent advantage can be an artefact of an incompetent control.** In a three-class source-attribution task, explicit binding appeared to win by 0.38 in Brier skill score against generic controls; once the control could use the information it was given, the difference vanished. Any advantage of agent-held evidence must survive a competent conventional competitor.
- **Local statistics can stand in for history.** The source of an episode was carried by the magnitude and variability of the updates within it, and resetting the state before every episode did not reduce decodability. Any claim that an agent uses its accumulated history must pass a reset criterion, because shortcuts available in the inputs will be found [19].

The second study [24] adds two cautions.

- **A model-centred quantity can be the optimizer's bookkeeping.** Under Adam the size of every step is set by moments the optimizer already keeps. In a language model, the accumulated norm of parameter change was reconstructed from the learning rate times the norm of the first moment with held-out R^2 0.94–0.96 (0.96–0.97 with the summed learning rate added), and a functional distance on probes was reconstructed from the change of loss on the same probes. Any candidate index must be reconstructed from free telemetry before its predictive value is tested: if it is reconstructed, its value cannot be identified.
- **An exception found by extrapolation must be re-tested inside its regime.** A functional distance seemed to escape reconstruction in one schedule family, with that family left out of training and one seed (R^2 0.81); on eight new seeds of that family, leaving one seed out, it was reconstructed at 0.99. An exception counts as evidence only once it recurs where it was found.

Physical components add a caution of their own: an advantage of a physical clock must be compared with a matched informational simulation of the same signal. Such a simulation controls for the statistics of the signal in non-adversarial tasks, under threat model A. Under threat model B what matters is only where the source resides relative to the checkpoint: a simulation inside the checkpoint is restored with it, and one outside it is a conventional trusted timer. Irreversibility counts only through the evidence it provides.

7. Risks and a Design Principle

Detecting one's own suspension, restoration, or copying is a form of situational awareness, which safety research measures because it can undermine evaluations [4, 29]; current models already distinguish evaluation from deployment transcripts well above chance [39]. An indicator of branching or rollback is also an indicator of being evaluated, since evaluations routinely branch and reset checkpoints. Rollback is, moreover, a legitimate instrument of governance: protocols that permit authorised rollback while preventing unauthorised rollback have been proposed precisely because recovery from corruption requires it [9], version control of agent memory lets

users roll it back on request in natural language [62], and the difficulty of restoring a self-modifying agent's behaviour by reverting its self-description is itself a governance problem [63].

We therefore propose one design principle: *access to evidence of restoration confers no control over it*. The evidence is read-only to the agent; the agent has no write access to the mechanism and no authority over the orchestrator that restores it, and operators can restore it whatever it concludes. The principle is the evidential counterpart of safe interruptibility and corrigibility, where an agent must learn neither to prevent nor to seek interruption [42], interruptions it can detect are pruned from what it learns [16], and an incentive to observe a variable is formally distinct from an incentive to control it [17]. Those results concern interventions to come; here the intervention has already happened, and the question is what the agent may do with evidence of it. Three kinds of access are distinguished: *bounded epistemic access*, in which the evidence informs uncertainty, estimates of staleness, invalidation of cached knowledge, and requests for verification; *policy access*, in which it may influence external actions directly; and *lifecycle authority*, the power to block, authorise, or alter restoration. The first is the default proposed here; the second is a separate capability and safety decision that requires mediation, audit, and revocable authorisation; the third is never granted to the agent. Privilege separation does not, however, make behaviour restorable by itself. By the proposition of §3, a restored agent that reads non-restorable evidence receives inputs different from those it received when its state was saved; if its policy conditions on them — by refusing work after a detected rollback, for instance — restoration reproduces the refusal rather than the earlier behaviour, and repeated restoration only widens the gap. Restoring behaviour therefore requires that discontinuity evidence enter only epistemic functions — estimates of staleness, invalidation of cached knowledge, calibrated uncertainty — rather than unconstrained action selection — readability by the self-model and readability by the whole policy are not the same thing — or that the platform virtualise the continuity evidence during governance recovery and evaluation. Virtualisation must itself be an authorised and audited operation, enforced and recorded by the trusted monitor rather than by the host [9], so that a compromised orchestrator cannot invoke it to mask a real discontinuity; it trades the protection the evidence offers under threat model B for operator control, and the trade should be explicit. Capabilities of this kind should be studied in contained settings, kept visible to overseers on the same terms as to the agent, and reported as dual-use.

Temporal self-models also touch properties discussed in the scientific assessment of consciousness in AI [11] and in the debate on AI welfare [32, 37], and research near them should follow principles for responsible conduct [10]. The proposal makes no claim in this direction, and the evidence counsels against one. Temporal continuity and consciousness dissociate: a professional musician with dense amnesia after encephalitis remained conscious while losing the continuity of experience [68]. The human capacity closest to the present target is autonoetic consciousness — awareness of oneself in subjective time [66] — and only its functional analogues are at issue here; the leading theories of consciousness, moreover, remain contested even under preregistered adversarial tests [13].

8. Conclusion

Biological organisms develop along trajectories that cannot be restored from a complete checkpoint, whereas artificial agents are commonly engineered so that their computational state

can be copied and reinstated. The difference is not that one exists in time and the other does not, but that in the latter, evidence of elapsed time and of change can be erased together with the saved state. The mechanisms that could preserve such evidence already exist — in hardware security, in cryptography, and in the neuroscience of timing — but they serve the platform and the guard checks of protected modules, not the agent's model of itself. We have proposed a classification of the evidence an agent can hold, an elementary result that locates the problem, a separation between a continuity anchor that detects discontinuities and a dual clock that serves the agent's model of its own change, a sequence of tests with the competitors that could close them, and a principle that separates access to evidence from control over restoration. One level has since been tested for its cheapest candidates, and the statistics the optimizer already keeps proved sufficient [24]; the accompanying exploratory study, which tests no level, had pointed the same way. The other tests, including those of the continuity anchor, where the proposal is most distinctive, are offered to others. Whether a time of their own makes persistent agents more capable, or only more complicated, is an empirical question, and the elements of a method for answering it are in place.

Declarations

Author contributions. Ioan Daniel Ionici originated the idea and the research question, determined the scope and the claims, made all editorial decisions, and takes full responsibility for the content of this article.

Use of generative artificial intelligence. Claude Opus 5.5 (Anthropic; model identifier claude-opus-5-5), used through the Claude desktop application in October and November 2026, assisted with formulating and criticising the argument, searching the literature, checking every bibliographic reference against Crossref, DataCite, arXiv, and publisher or proceedings records, drafting the text in English, and copy-editing it. Microsoft Copilot and DeepSeek were consulted on earlier formulations of the proposal, and drafts of this article were reviewed by Microsoft Copilot and by the Google DeepReview system. Version 1.2 was prepared in the same way; the omitted prior work it cites was identified during AI reviews of a follow-up research protocol and an AI-assisted priority search (Google Argon, Microsoft Copilot). Version 1.3 revises the closure rule, the description of the accompanying study's role, and the statement of priority in response to an automated review by the CMU Paper Reviewer (GPT-5.5). Version 1.4 reports the second study, carried out with the same assistant under the author's direction [24], and was prepared in the same way. The author reviewed all AI outputs and made every decision about what to adopt. No AI system is listed as an author, because none can take responsibility for the work; responsibility for the content, including any errors, rests with the author.

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